Methods 02: Set up Python and VS Code

MPhil Past and Future Climates · Computational techniques for paleoclimate research · Francesco Muschitiello, University of Cambridge

Before the practical you need three things: the latest course files, an updated Python toolbox, and VS Code set up to run the notebook with that toolbox. This takes about an hour the first time, mostly waiting for downloads. You do not need to understand the code to do it.

What you need and why

> ### Conda environment
>
> A separate toolbox of Python packages for one project. This practical needs a few special packages (pyleoclim, pylipd, xeofs). Keeping them in their own toolbox, called `pfc`, means they do not clash with your other work.

> ### Kernel
>
> The engine that runs the code in a notebook. The notebook itself is only a document. VS Code must be told to use the engine from the `pfc` toolbox, otherwise the special packages are missing.

Step	What you do	Terminal needed?
1–3	Get the files, update the toolbox, test it	Yes (copy and paste three short commands)
4–6	Install VS Code, open the folder, choose the kernel	No
7–8	Run the notebook and work on your own copy	No

Open a terminal: on a Mac press `Cmd+Space`, type **Terminal** and press Enter. On Windows open **Anaconda Prompt** (or **Miniforge Prompt**, if that is what you installed) from the Start menu.

Before you start: is conda installed?

Open a terminal (see above) and type `conda --version`, then press Enter.

- If it prints a version number (for example `conda 24.5.0`), conda is installed. Go on to step 1.
- If you see **command not found** (or **conda is not recognized** on Windows), install conda first. Go to **github.com/conda-forge/miniforge**, download the installer for your computer and follow the instructions on that page, accepting the default options. Then close the terminal, open a new one (on Windows: **Miniforge Prompt**) and type `conda --version` again.

1. Get the materials

If you cloned the course repository with git, go to the repository folder in the terminal and run:

```
git pull
```

First time, or no git? Open the course repository page on GitHub, click the green **Code** button, choose **Download ZIP**, and unzip it. Put the unzipped folder somewhere simple, such as a folder called

`PastFutureClimates` in your home folder. On a Mac, avoid the Desktop, Documents and Downloads folders, because macOS can block VS Code from reading them (see Troubleshooting).

If you downloaded the ZIP last time, download the ZIP again from the course repository and unzip it. If `git pull` complains about local changes, you edited an original file last time: ask the instructor before you continue. The **repository folder** is the folder that contains `environment.yml`.

2. Update the environment

This adds the three new packages to your `pfc` toolbox. In the terminal, from the repository folder, run:

```
conda env update -f environment.yml
conda activate pfc
```

It can take 10 minutes or more. Wait until the prompt comes back.

> **If you see an error saying the environment `pfc` does not exist,** you never created it. Create it from the file instead, then activate it:
>
> ```
> conda env create -f environment.yml
> conda activate pfc
> ```

> **If the update reports conflicts or errors,** make a separate new toolbox instead. This always works and does not touch `pfc`:
>
> ```
> conda env create -f environment.yml -n pfc_methods2
> conda activate pfc_methods2
> ```
>
> Wherever this guide says `pfc`, use `pfc_methods2` instead.

3. Test the installation

With the environment active, run:

```
cd practicals/02_methods
python check_setup.py
```

The **last line must say `ALL OK`**. Run it while you have internet: the first time it downloads the coastline files used for maps. If the last line says `NOT READY`, copy the whole output and send it to the instructor.

4. Install VS Code and two extensions

1. Download and install Visual Studio Code from **code.visualstudio.com**.
2. Open VS Code and click the **Extensions** icon in the left bar (four small squares), or press `Cmd+Shift+X` (Windows: `Ctrl+Shift+X`).
3. Search for **Python** (publisher: Microsoft) and click **Install**.
4. Search for **Jupyter** (publisher: Microsoft) and click **Install**.

5. Open the repository folder and trust it

1. Choose **File > Open Folder** and select the **repository folder** (the one with `environment.yml`). Open the folder, not just the notebook file.
2. VS Code asks whether you trust the authors of the files in this folder. Click **Yes, I trust the authors**.

> **Why trust matters.** If you do not trust the folder, VS Code opens it in **Restricted Mode**, which blocks the images in the notebook and the Python tools. To fix it later, click **Restricted Mode** in the bottom-left corner and choose to trust the folder.

6. Open the notebook and choose the kernel

1. In the file list on the left, open `practicals/02_methods`, then click `Methods_02_Paleoclimate_Computational_Techniques.ipynb`.
2. At the top right of the notebook click **Select Kernel**, then **Python Environments**.
3. Choose **pfc (Python 3.11)**. The name of the kernel now shows at the top right.

If pfc is not in the list

1. Reload the window: press `Cmd+Shift+P` (Windows: `Ctrl+Shift+P`), type **Developer: Reload Window** and press Enter. Then try Select Kernel again.
2. Still missing? Press `Cmd+Shift+P` (Windows: `Ctrl+Shift+P`), run **Python: Select Interpreter**, choose **Enter interpreter path…**, then **Find…**, and browse to the Python program inside the `pfc` environment:
   - **Mac:** `<conda folder>/envs/pfc/bin/python`, for example `/opt/anaconda3/envs/pfc/bin/python` or `~/miniforge3/envs/pfc/bin/python`.
   - **Windows** (same steps, not tested by the instructor): `<conda folder>\envs\pfc\python.exe`.
3. Then click **Select Kernel** again and choose `pfc`.

> **Tip: find the exact path.** In the terminal, with `pfc` active, run `python -c "import sys; print(sys.executable)"`. It prints the full path to copy.
>
> **Tip for the Mac file dialog:** press `Cmd+Shift+G` to type a path. The go-to box only suggests folders: type the folder path (for example `/opt/anaconda3/envs/pfc/bin`), press Enter, then pick the file named `python`.

7. Run the notebook

- Click into a cell and press `Shift+Enter` to run it and move to the next one. Run the cells from top to bottom.
- Or click **Run All** in the toolbar at the top of the notebook.
- The cell that loads the coral files takes about **25–30 seconds**. That is normal.
- A finished cell shows a green tick and the time it took on its left.

8. Work on your own copy

Do not edit the original notebooks. In the file list, right-click the notebook, choose **Copy**, then right-click the folder and choose **Paste**. Rename the copy (right-click > **Rename**), for example `Methods_02_yourname.ipynb`, and work in it. That way `git pull` can update the originals without clashing with your changes.

Backup route: JupyterLab in your browser

If VS Code will not cooperate, use JupyterLab. It runs the same notebook in your web browser. In the terminal, starting from the **repository folder** (if you already ran step 3 you are inside `practicals/02_methods`, so skip the `cd` line):

```
conda activate pfc
cd practicals/02_methods
jupyter lab
```

A browser tab opens. Double-click the notebook, and if asked for a kernel pick **Python (pfc)** or the default Python 3 kernel. Keep the terminal open while you work; close it when you are finished.

Troubleshooting

Problem	What to do
`ModuleNotFoundError` (pyleoclim, pylipd or xeofs)	Wrong kernel. Click the kernel name at the top right and choose `pfc` (step 6).
`pfc` is not in the kernel list	Reload the window, or use **Python: Select Interpreter > Enter interpreter path** (step 6).
Images missing, or the kernel cannot start	The folder is in Restricted Mode. Click **Restricted Mode** (bottom left) and trust the folder.
Mac: `PermissionError: Operation not permitted` when the notebook reads data	macOS is blocking VS Code. Open **System Settings > Privacy & Security > Files and Folders**, find **Visual Studio Code** and switch on the folder where the repository lives (Desktop, Documents or Downloads), or give it **Full Disk Access**. Quit VS Code with `Cmd+Q` and reopen it. Or move the repository to a folder outside those three.
A map cell fails with a connection error	Cartopy needs the internet once to download coastlines. Run `python check_setup.py` while online.
The notebook says to open it from `practicals/02_methods`	Open the **repository folder** in VS Code (step 5) and open the notebook from the file list.
The environment update seems stuck for more than 20 minutes	Press Ctrl+C to stop it and ask the instructor. Older Anaconda installations solve environments slowly; Miniforge (recommended in the repository README) is much faster.
The first cell is slow	Normal. Loading the libraries and data can take up to a minute the first time.

Anything else	Copy the **whole** error message and post it in the course channel, saying which operating system you use.

Before the practical

- ☐ `git pull` done, or the new ZIP downloaded
- ☐ Environment updated (`pfc`, or `pfc_methods2`)
- ☐ `python check_setup.py` ends with `ALL OK`
- ☐ VS Code installed with the Python and Jupyter extensions
- ☐ Repository folder opened and trusted; notebook runs with the `pfc` kernel
- ☐ Your own copy of the notebook made, and two questions written down for class

Instructions checked on 19 October 2026 and may change as VS Code and its extensions update. Windows paths follow the same pattern as on a Mac but have not been tested by the instructor.